Supplementation of O_2-containing gas nanobubble water to enhance methane production from anaerobic digestion of cellulose

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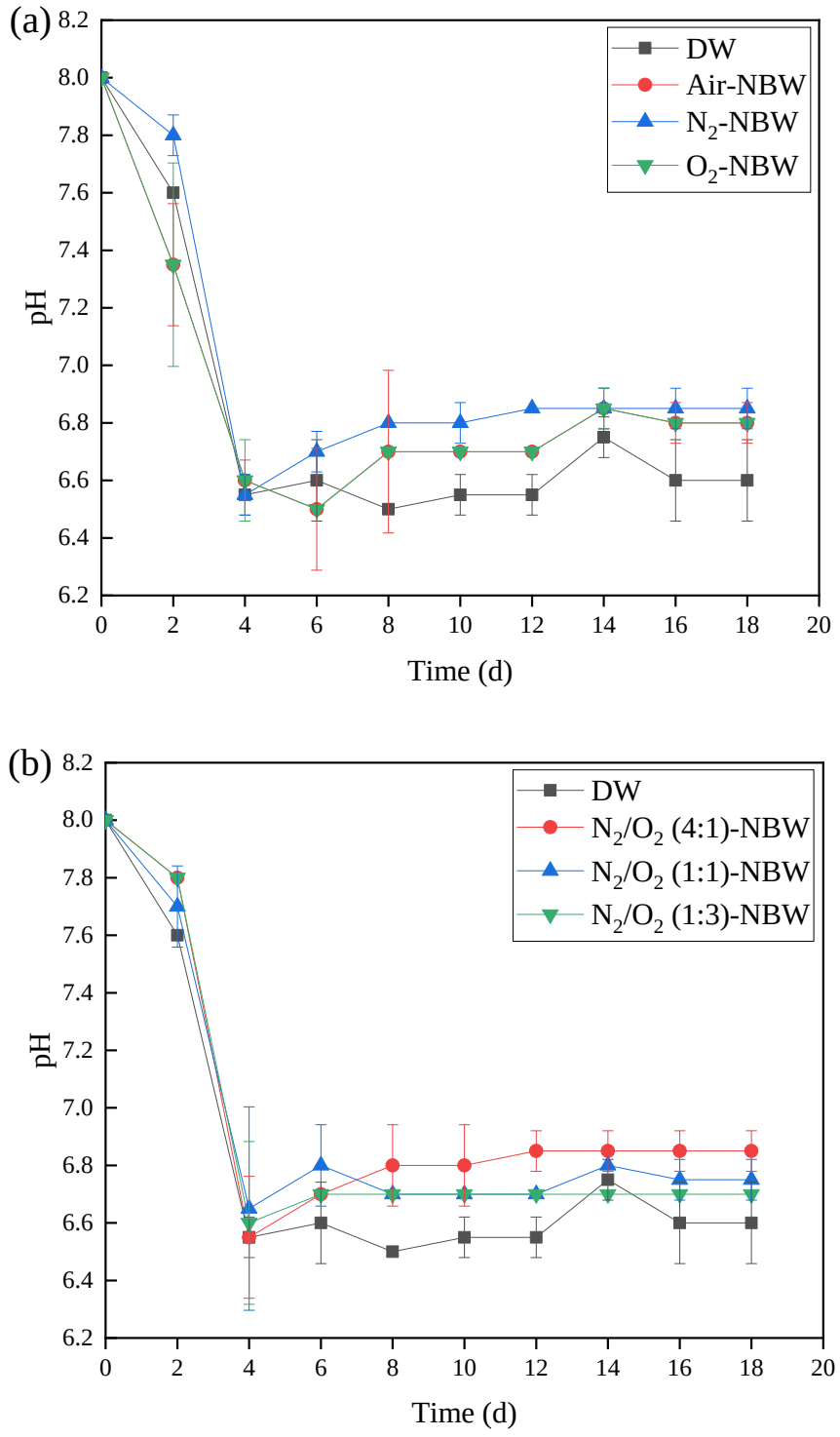


Fig. S1. Changes in pH value during 18 days' AD of cellulose with DW and NBW addition. (a) Air-NBW, N_2-NBW and O_2-NBW addition, and (b) N_2/O_2 (4:1)-NBW, N_2/O_2 (1:1)-NBW and N_2/O_2 (1:3)-NBW addition. DW-deionized water, NBW-nanobubble water.

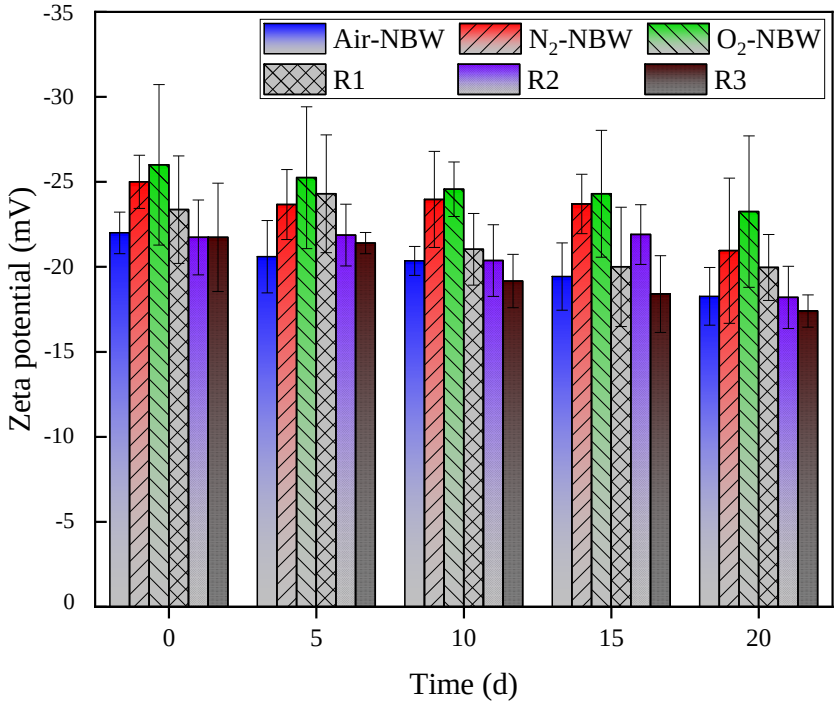


Fig. S2. Changes in zeta potential of different NBW over 20 days' storage at 35±1°C. N_2/O_2 (4:1)-NBW, N_2/O_2 (1:1)-NBW and N_2/O_2 (1:3)-NBW are denoted as R1, R2 and R3, respectively. NBW-nanobubble water.

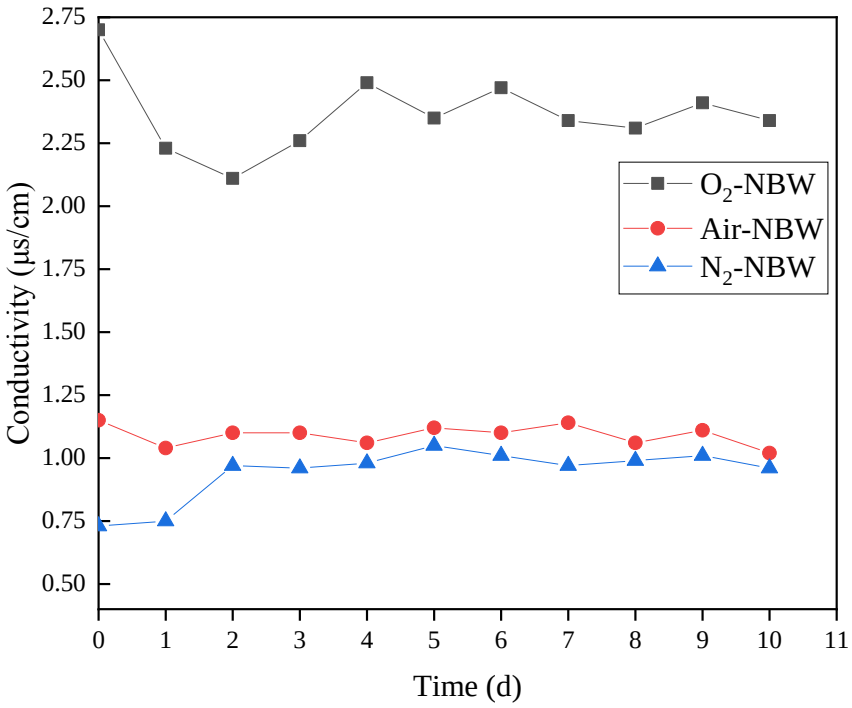


Fig. S3. Changes in conductivity of O_2-NBW, Air-NBW and N_2-NBW during 10 days' storage at 35±1°C. NBW-nanobubble water.

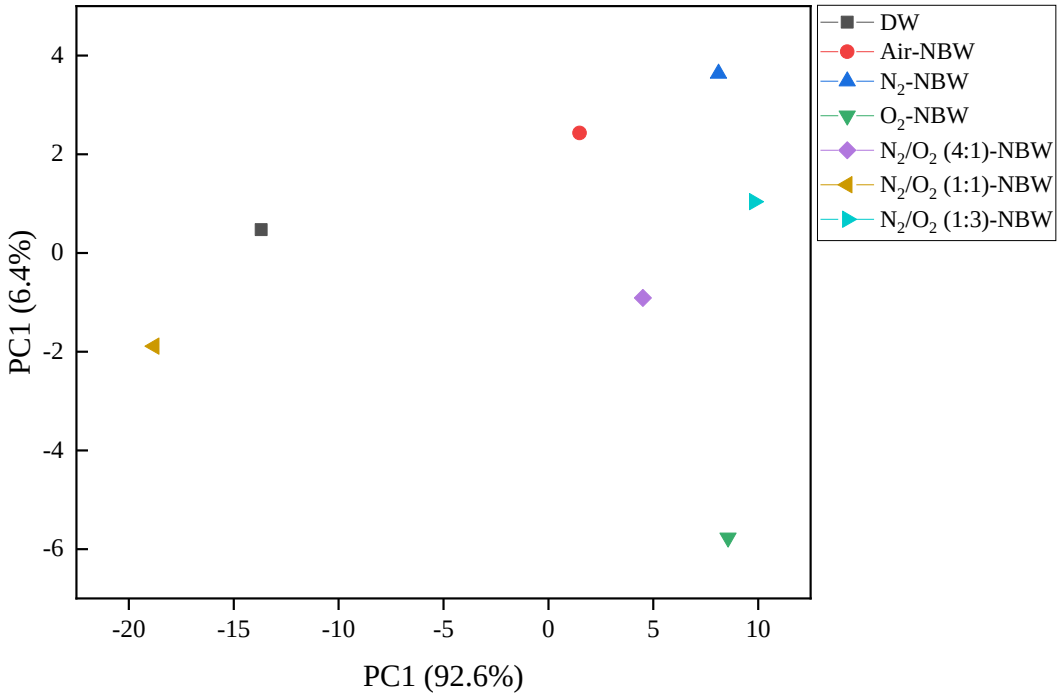


Fig. S4. Principal component analysis of the effects of NBW on the microbial consortia. DW-deionized water, NBW-nanobubble water.

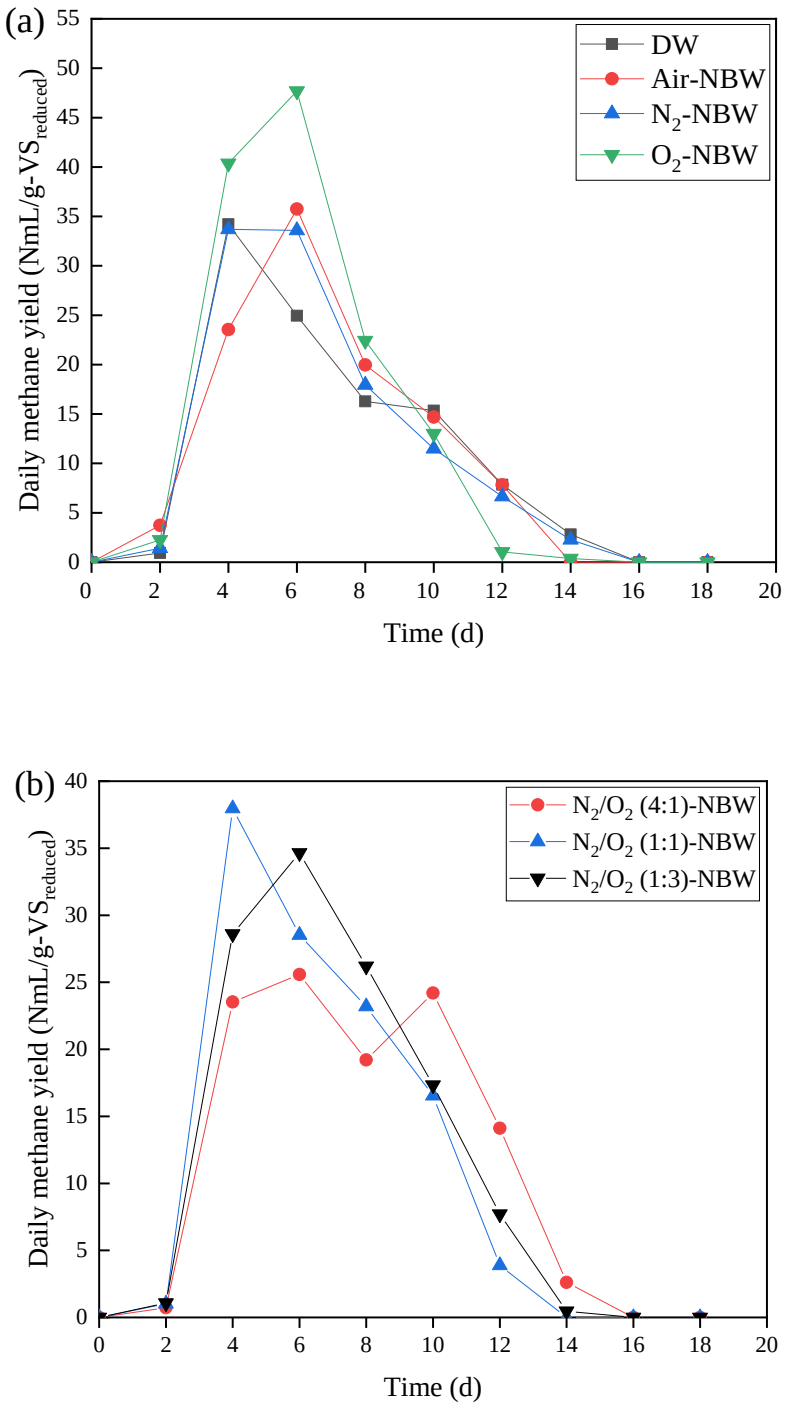


Fig. S5. Daily methane yield during the anaerobic digestion supplemented with DW and different NBW. (a) Air-NBW, N_2-NBW and O_2-NBW addition, and (b) N_2/O_2 (4:1)-NBW, N_2/O_2 (1:1)-NBW and N_2/O_2 (1:3)-NBW addition. DW-deionized water, NBW-nanobubble water.

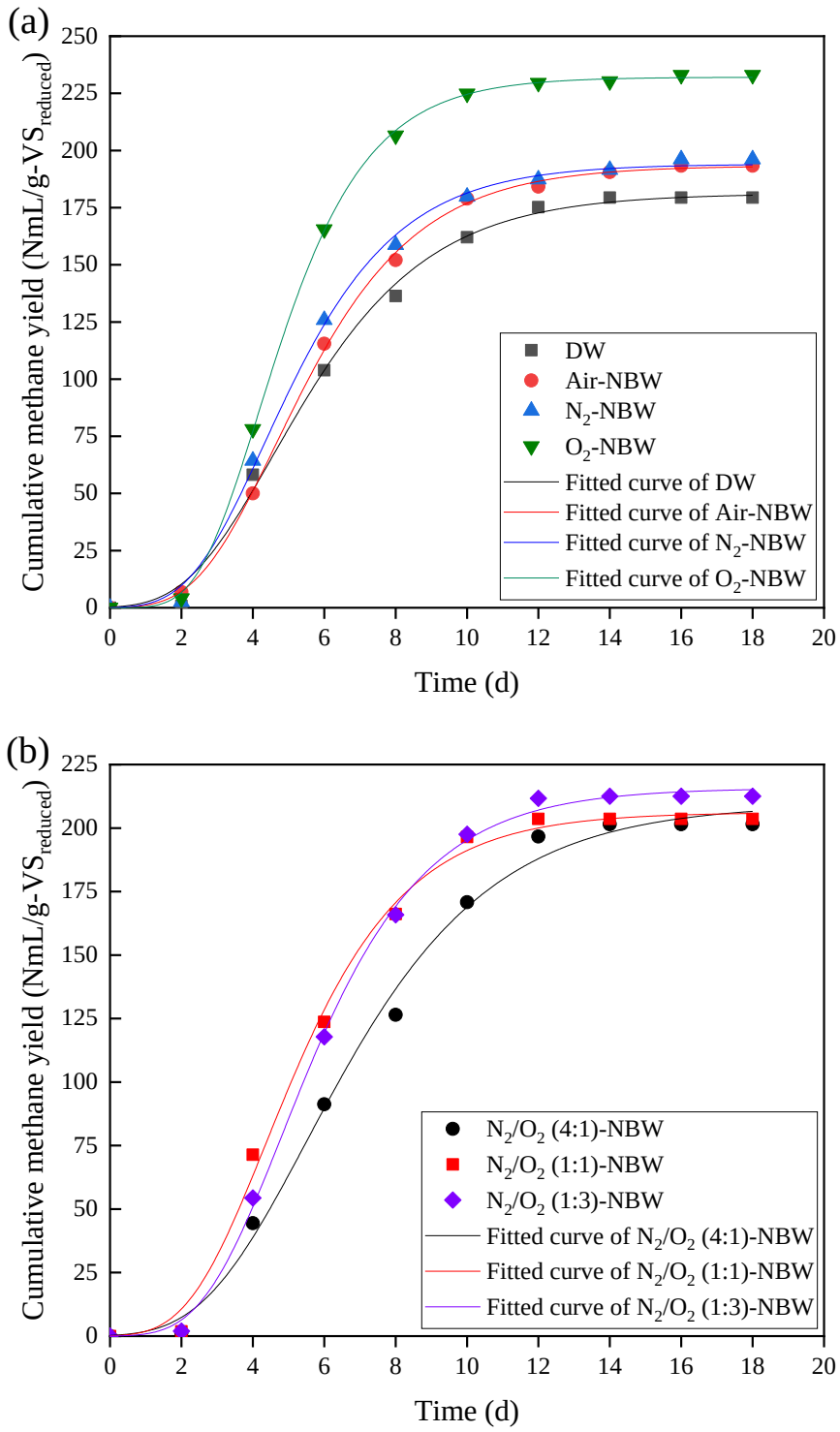


Fig. S6. Experimental and simulated results on cumulative methane production from the DW and NBW reactors. Air-NBW, N_2-NBW and O_2-NBW (a), N_2/O_2 (4:1)-NBW, N_2/O_2 (1:1)-NBW and N_2/O_2 (1:3)-NBW (b) by the modified Gompertz model. DW-deionized water, NBW-nanobubble water.